

Challenge: AWSRaid Lab

Platform: SOC Analyst Tier 1

Category: Cloud Forensics

Tactics: Persistence, Privilege Escalation, Credential Access

Tools: Splunk

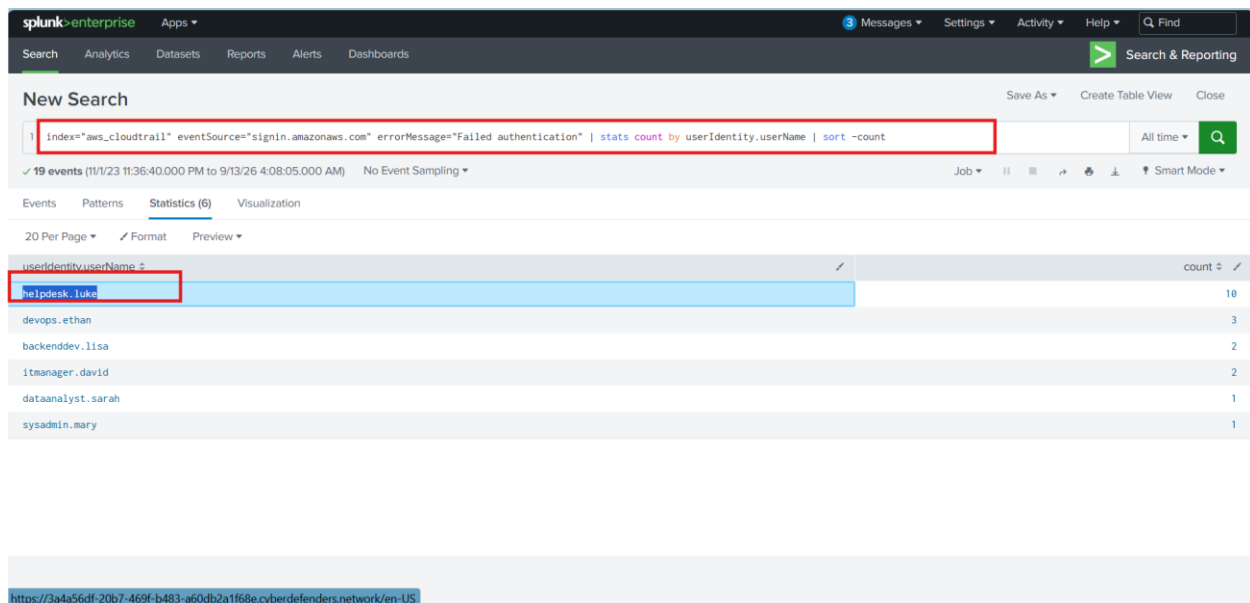
VIEWS LAB: [link](#)

Kịch bản: Investigate AWS CloudTrail logs using Splunk to identify unauthorized access, analyze configuration changes, and detect persistence mechanisms.

Q1

Knowing which user account was compromised is essential for understanding the attacker's initial entry point into the environment. What is the username of the compromised user?

➔ Trên thanh tìm kiếm: `index="aws_cloudtrail" eventSource="signin.amazonaws.com" errorMessage="Failed authentication" | stats count by userIdentity.userName | sort -count` (SPL)



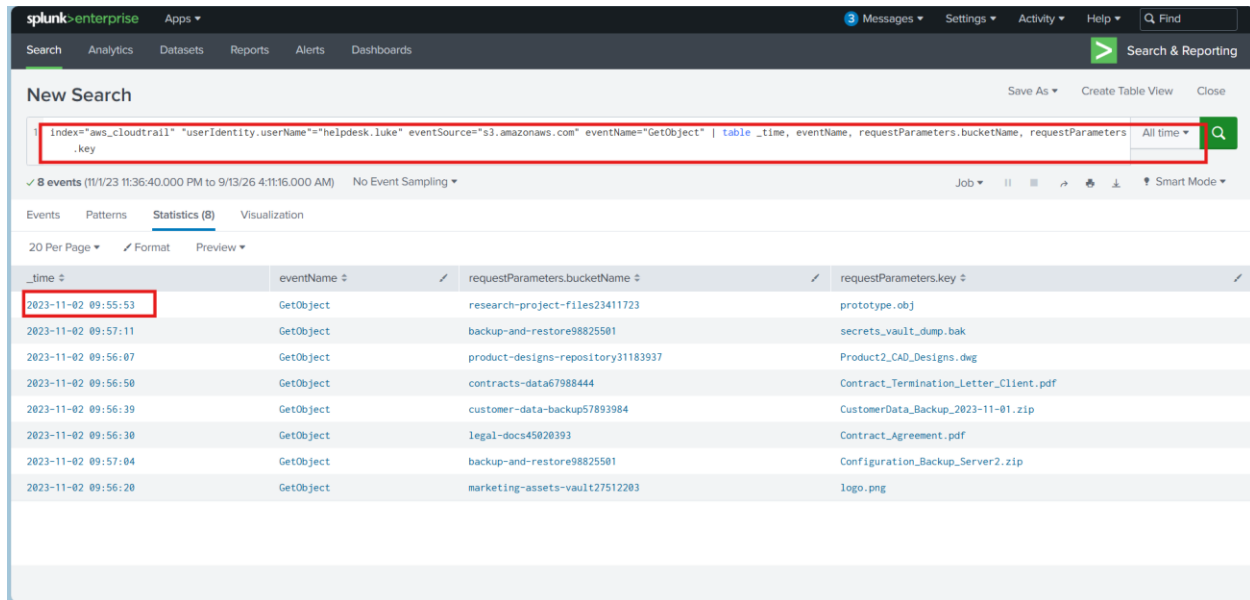
The screenshot shows the Splunk Enterprise search interface. The search bar contains the query: `index="aws_cloudtrail" eventSource="signin.amazonaws.com" errorMessage="Failed authentication" | stats count by userIdentity.userName | sort -count`. The search results are displayed in a table with two columns: `userIdentity.userName` and `count`. The results are sorted by count in descending order.

userIdentity.userName	count
helpdesk.luke	10
devops.ethan	3
backenddev.lisa	2
itmanager.david	2
dataanalyst.sarah	1
sysadmin.mary	1

Q2

We must investigate the events following the initial compromise to understand the attacker's motives. What is the timestamp for the first access to an S3 object by the attacker?

➔ `index="aws_cloudtrail" "userIdentity.userName"="helpdesk.luke" eventSource="s3.amazonaws.com" eventName="GetObject" | table _time, eventName, requestParameters.bucketName, requestParameters.key`



The screenshot shows the Splunk Enterprise interface with a search bar containing the query: `index="aws_cloudtrail" "userIdentity.userName"="helpdesk.luke" eventSource="s3.amazonaws.com" eventName="GetObject" | table _time, eventName, requestParameters.bucketName, requestParameters.key`. The search results are displayed in a table with 8 events. The first event is highlighted with a red box, showing the timestamp 2023-11-02 09:55:53.

_time	eventName	requestParameters.bucketName	requestParameters.key
2023-11-02 09:55:53	GetObject	research-project-files23411723	prototype.obj
2023-11-02 09:57:11	GetObject	backup-and-restore98825501	secrets_vault_dump.bak
2023-11-02 09:56:07	GetObject	product-designs-repository31183937	Product2_CAD_Designs.dwg
2023-11-02 09:56:50	GetObject	contracts-data67988444	Contract_Termination_Letter_Client.pdf
2023-11-02 09:56:39	GetObject	customer-data-backup57893984	CustomerData_Backup_2023-11-01.zip
2023-11-02 09:56:30	GetObject	legal-docs45020393	Contract_Agreement.pdf
2023-11-02 09:57:04	GetObject	backup-and-restore98825501	Configuration_Backup_Server2.zip
2023-11-02 09:56:20	GetObject	marketing-assets-vault27512203	logo.png

Q3

Among the S3 buckets accessed by the attacker, one contains a DWG file. What is the name of this bucket?

➔ `index="aws_cloudtrail" "userIdentity.userName"="helpdesk.luke" eventSource="s3.amazonaws.com" eventName="GetObject" "*.dwg" | table _time, requestParameters.bucketName, requestParameters.key`

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

1 `index="aws_cloudtrail" "userIdentity.userName"="helpdesk.luke" eventSource="s3.amazonaws.com" eventName="GetObject" "*" ".dwg" | table _time, requestParameters.bucketName, requestParameters.key` All time Q

✓ 1 event (11/1/23 11:36:40.000 PM to 9/13/26 4:13:32.000 AM) No Event Sampling Job II III IV V VI VII VIII Smart Mode

Events Patterns **Statistics (1)** Visualization

20 Per Page Format Preview

_time	requestParameters.bucketName	requestParameters.key
2023-11-02 09:56:07	product-designs-repository31183937	Product2_CAD_Designs.dwg

Q4

We've identified changes to a bucket's configuration that allowed public access, a significant security concern. What is the name of this particular S3 bucket?

➔ `index="aws_cloudtrail" "userIdentity.userName"="helpdesk.luke" eventName="PutBucketPublicAccessBlock" | table _time, requestParameters.bucketName, userIdentity.arn`

splunk enterprise Apps Messages Settings Activity Help Find

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New Search

Save As Create Table View Close

1 `index="aws_cloudtrail" "userIdentity.userName"="helpdesk.luke" eventName="PutBucketPublicAccessBlock" | table _time, requestParameters.bucketName, userIdentity.arn` All time Q

✓ 1 event (11/1/23 11:36:40.000 PM to 9/13/26 4:15:44.000 AM) No Event Sampling Job II III IV V VI VII VIII Smart Mode

Events Patterns **Statistics (1)** Visualization

20 Per Page Format Preview

_time	requestParameters.bucketName	userIdentity.arn
2023-11-02 09:58:01	backup-and-restore98825501	arn:aws:iam::141573590337:user/helpdesk.luke

Q5

Creating a new user account is a common tactic attackers use to establish persistence in a compromised environment. What is the username of the account created by the attacker?

New Search

1 index="aws_cloudtrail" "*CreateUser*" All time

✓ 1 event (11/23 11:36:40.000 PM to 9/13/26 4:21:23.000 AM) No Event Sampling

Events (1) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 month per column

List Format 20 Per Page

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1	INTERESTING FIELDS a awsRegion 1 # date_hour 1 # date_mday 1 # date_minute 1 a date_month 1 # date_second 1 a date_wday 1 # date_year 1 # date_zone 1 a eventCategory 1 a eventID 1 a eventName 1	>	11/2/23 9:59:33.000 AM	<pre>{ [-] awsRegion: us-east-1 eventCategory: Management eventID: 88be7234-8f30-4568-9c71-96df43d89870 eventName: CreateUser eventSource: iam.amazonaws.com eventTime: 2023-11-02T09:59:33Z eventType: AwsApiCall eventVersion: 1.08 managementEvent: true readOnly: false recipientAccountId: 141573590337 requestID: 3660219b-2197-4e42-8f4b-15021a642bee requestParameters: { [+]} responseElements: { [+]} sessionCredentialFromConsole: true sourceIPAddress: 185.192.70.78 userAgent: AWS Internal }</pre>

3a4a56df-20b7-469f-b483-a60db2a1f68e.cyberdefenders.network/.../search?q=...

List Format 20 Per Page

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1	INTERESTING FIELDS a awsRegion 1 # date_hour 1 # date_mday 1 # date_minute 1 a date_month 1 # date_second 1 a date_wday 1 # date_year 1 # date_zone 1 a eventCategory 1 a eventID 1 a eventName 1 a eventSource 1 a eventTime 1 a eventType 1 # eventVersion 1 a index 1 # linecount 1 a managementEvent 1 a punct 1 a readOnly 1 # recipientAccountid 1 a requestID 1	>	11/2/23 9:59:33.000 AM	<pre>{ [-] awsRegion: us-east-1 eventCategory: Management eventID: 88be7234-8f30-4568-9c71-96df43d89870 eventName: CreateUser eventSource: iam.amazonaws.com eventTime: 2023-11-02T09:59:33Z eventType: AwsApiCall eventVersion: 1.08 managementEvent: true readOnly: false recipientAccountId: 141573590337 requestID: 3660219b-2197-4e42-8f4b-15021a642bee requestParameters: { [+]} responseElements: { [+]} sessionCredentialFromConsole: true sourceIPAddress: 185.192.70.78 userAgent: AWS Internal userIdentity: { [+]} }</pre>

Show as raw text

host = splunk | source = s3://aws-cloudtrail-logs-141573590337-619792c3/AWSLogs/141573590337/Clou... | sourcetype = aws:cloudtrail

List ▾ Format 20 Per Page ▾		
< Hide Fields ▮ All Fields	i Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1 INTERESTING FIELDS # awsRegion 1 # date_hour 1 # date_mday 1 # date_minute 1 # date_month 1 # date_second 1 # date_wday 1 # date_year 1 # date_zone 1 a eventCategory 1 a eventID 1 a eventName 1 a eventSource 1 a eventTime 1 a eventType 1 # eventVersion 1 a index 1 # linecount 1 a managementEvent 1 a punct 1 a readOnly 1 # recipientAccountid 1 a requestID 1	> 11/2/23 9:59:33.000 AM	<pre>{ "eventVersion": "1.08", "userIdentity": { "type": "IAMUser", "principalId": "AIDASB5TRRVAQAFZZR6W", "arn": "arn:aws:iam:141573590337:user/helpdesk.luke", "accountId": "141573590337", "accessKeyId": "ASIASB5TRRVAZSE73G7", "userName": "helpdesk.luke", "sessionContext": { "sessionIssuer": {}, "webIdentityFederationData": {}, "attributes": { "creationDate": "2023-11-02T09:54:04Z", "mfaAuthenticated": "false" } }, "eventTime": "2023-11-02T09:59:33Z", "eventSource": "iam.amazonaws.com", "eventName": "CreateUser", "awsRegion": "us-east-1", "sourceIPAddress": "185.192.70.78", "userAgent": "AWS Internal", "requestParameters": { "userName": "marketing.mark", "responseElements": { "user": { "path": "/", "userName": "marketing.mark", "userId": "AIDASB5TRRVAUL5IB4BC4", "arn": "arn:aws:iam:141573590337:user/marketing.mark", "createDate": "Nov 2, 2023 9:59:33 AM" } } }, "requestID": "3660219b-2197-4e42-8f4b-15021a642bee", "eventID": "88be7234-8f30-4568-9c71-96df43d89878", "readOnly": false, "eventType": "AwsApiCall", "managementEvent": true, "recipientAccountid": "141573590337", "eventCategory": "Management", "sessionCredentialFromConsole": "true" }</pre> Show syntax highlighted host = splunk source = s3://aws-cloudtrail-logs-141573590337-619792c3/AWSLogs/141573590337/Clou... sourcetype = aws:cloudtrail

Q6

Following account creation, the attacker added the account to a specific group. What is the name of the group to which the account was added?

List ▾ Format 20 Per Page ▾		
< Hide Fields ▮ All Fields	i Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1 INTERESTING FIELDS # awsRegion 1 # date_hour 1 # date_mday 1 # date_minute 1 # date_month 1 # date_second 1 # date_wday 1 # date_year 1 # date_zone 1 a eventCategory 1 a eventID 1 a eventName 1 a eventSource 1 a eventTime 1 a eventType 1 # eventVersion 1 a index 1 # linecount 1 a managementEvent 1 a punct 1 a readOnly 1 # recipientAccountid 1 a requestID 1	> 11/2/23 9:59:38.000 AM	<pre>{ "eventVersion": "1.08", "userIdentity": { "type": "IAMUser", "principalId": "AIDASB5TRRVAQAFZZR6W", "arn": "arn:aws:iam:141573590337:user/helpdesk.luke", "accountId": "141573590337", "accessKeyId": "ASIASB5TRRVAZSE73G7", "userName": "helpdesk.luke", "sessionContext": { "sessionIssuer": {}, "webIdentityFederationData": {}, "attributes": { "creationDate": "2023-11-02T09:54:04Z", "mfaAuthenticated": "false" } }, "eventTime": "2023-11-02T09:59:38Z", "eventSource": "iam.amazonaws.com", "eventName": "AddUserToGroup", "awsRegion": "us-east-1", "sourceIPAddress": "185.192.70.78", "userAgent": "AWS Internal", "requestParameters": { "groupName": "AWSIITR", "userName": "marketing.mark", "responseElements": null }, "requestID": "7a8ed069-cc29-46e8-b3fc-ba6387155629", "eventID": "4c47cd82-28c1-4ae7-9ad6-78aa5232d67b", "readOnly": false, "eventType": "AwsApiCall", "managementEvent": true, "recipientAccountid": "141573590337", "eventCategory": "Management", "sessionCredentialFromConsole": "true" }</pre> Show syntax highlighted host = splunk source = s3://aws-cloudtrail-logs-141573590337-619792c3/AWSLogs/141573590337/Clou... sourcetype = aws:cloudtrail